PCA → Principal Component Analysis Technique

→ Unsupervised M.L

→ Reducing the No of column

100 column in data → 30 - 40

→ Dimensionality Reduction Technique

→ Converts Co-related Columns to new col.

Principle : Covariance → relationship b/w the columns

Feature Extraction ✓ Feature Creation Create new ↓ based on exist

DOB 07-01-24 | Age → FE 3

Weight | height | BMI

x_1	x_2	x_3	x_4	x_5	⇒	PC_1	PC_2
54.65	76.12	81.34	12.75	8.62		44.64	16.12

F_1 F_2 F_3 F_4 F_5 ⟶ 100% Info

PC_1	PC_2	PC_3	PC_4	PC_5	
60%	20%	10%	6%	4%	EVR

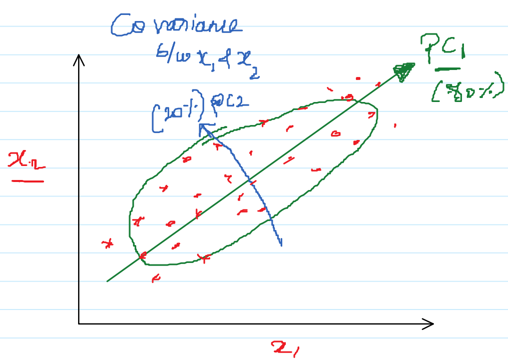
PC_1 PC_2 ⟶ 80% → info → Sufficient

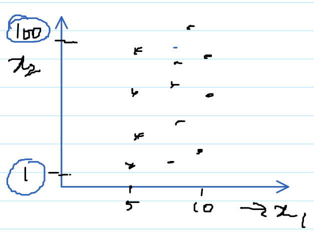
PC_1 → highest Info/EVR

PC_2 → Second highest EVR

x_1	x_2

data ⇒ $x_1 \rightarrow 80\%$





x_1 → more Vari ?

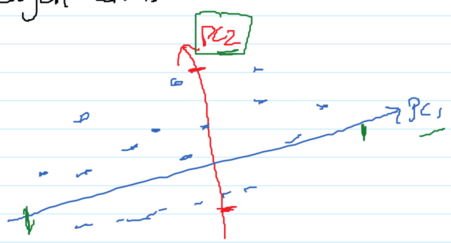
x_2 ⇒ more Variance!!? ✓

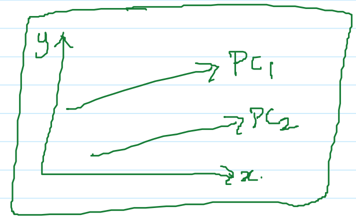
x_2 is more

x_2 alone...

Variance = Eigne values

direction ⇒ Eigen Vectors





There is a relation

$\perp r$

Relation

EVR ⇒ Explained Variance Ration

Assume 3 Features

Variance

Model ⇒ $PC_1 = 40$

$PC_2 = 20$

$PC_3 = 5$

$$EVR = \frac{\text{Variance of P.C}}{\text{total variance}}$$

$$EVR \text{ of } PC_1 = \frac{\text{Variance of } PC_1}{\text{total var}} = \frac{40}{40+20+5} = 61.45\%$$

$F_1 \; F_2 \; F_3 \rightarrow 100\%$ info

$$EVR \text{ of } PC_2 = \frac{20}{65} = 30.76\%$$

$\boxed{PC_1 \; PC_2}$ $\; P_3 \Rightarrow PC_1 + PC_2$

↓ 61.54% ↓ 30.7% ↓ 7.7% ↓ 90.56%

$$EVR \text{ of } PC_3 = \frac{5}{65} = 7.7\%$$

3 Feature ⇒ 2 Features

PC_1	PC_2	PC_3	P_4	P_5	P_6	---	P_{20}
↓	↓	↓	↓				
30%	25%	10%	5%	-	---		

Cummulative

$PC_1 + P_2 = 55\%$

$P_4 + PC_2 + PC_3 = 65\%$

$PC_1 + PC_2 + P_3 + P_4 = 75\%$

20 → only 5 col

$PC_1 + \text{--} \; PC_5 = 90\%$

Step by step

1. Standardization → Scaling
2. find Co-Variance Matrix

3. find Eigen value & Eigen Vector
4. Sort the PCs based EVR
5. Select the PC so that it can give max info
6. Transform the data

CROSS Validation

→ K-fold Cross validation

Test data

I F_1 200 | 200 | | | F_5 — $F_1 F_2 F_3 F_4$ Train, F_5 Test

II F_1 Test — $F_2 F_3 F_4 F_5$ Train, F_1 Test

III F_2

F_3

F_4

Limitation

→ classification — Unbalanced → K-Cross Validation Fails

1000 → Yes 800, NO 200

F_1 200 → 100 Y, 100 N; F_2 200 → 200 Yes, 0 N; F_3 200 → 200 NO, 0 Yes; F_4 200; F_5 200 → 200 Yes, 0 NO

Test → Model biased

2. Stratified

Data clean, FE, Data line

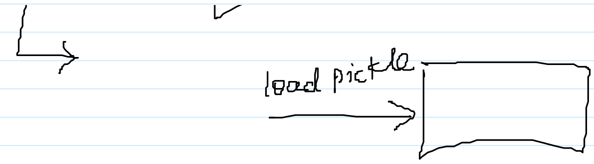
10,000 rec → Null + Outl + Skew_data + encod + FE + Scaling → save → pickle

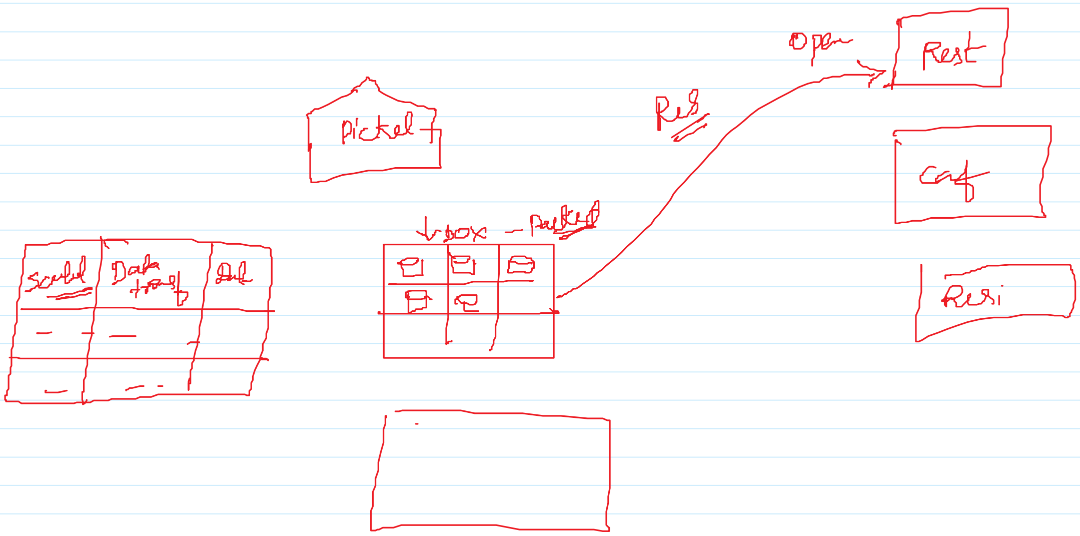
model → prediction

Prediction of disease

10,000 rec → model

Ask → new testa


load pickle


pickel f
box - packed
Res
open
Rest
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Data